Chemistry-Aware Machine Translation Evaluation with Domain Terminology Verification

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Abstract

Chemical translation evaluation must distinguish general quality from retention of technical terminology. We describe a segment-level methodology combining Google Patents Chemical abstract pairs with an article-focused JRC-Acquis legal comparison corpus. The pipeline constructs source snapshots, extracts reference-side terminology using complementary linguistic and learned approaches, attaches external evidence, and selects exact reference spans through domain-specific large language model curation. Translation generation separates a source-only baseline from an explicitly reference-derived, oracle terminology condition. Evaluation combines corpus lexical overlap, segment-level learned similarity, and strict and variant-aware term coverage. Intermediate artifacts and provenance support controlled comparisons. Benchmark preparation is ongoing; comparative translation results are not yet available. The protocol therefore makes no claim of improved MT quality or chemically validated annotations.

1 Introduction

Chemical translation requires preserving distinctions in names, materials, processes, formulas, and quantities. General MT metrics assess overlap or learned similarity [1, 2, 3, 4]; these differ from checking whether a specific technical term is retained correctly. We investigate how reference-side terminology annotations complement conventional evaluation, and how providing such terminology affects a controlled translation comparison.

The primary domain is chemistry patent abstracts. JRC-Acquis legal articles offer a distinct, terminology-sensitive comparison domain, not additional chemical training data. The evaluation unit is a bounded segment, potentially containing several sentences, rather than a complete patent or long legal document. The intended scale is a few hundred tokens; actual corpus lengths are reported instead of assuming a shared 256-token mean.

The contributions are multilingual benchmark construction, provenance-preserving terminology extraction and curation, and an evaluation protocol separating general quality from term retention. These preserve the chemistry-aware evaluation objective. The infrastructure is implemented, but the full translation comparison remains planned; no model ranking or superiority claim is made.

2 Related Work

Large language models (LLMs) have been investigated as translation systems through studies of translation quality, prompting, and adaptation. Hendy et al. compare GPT models with research and commercial systems, reporting competitive high-resource performance but limitations for low-resource languages in their evaluation [5]. Vilar et al. demonstrate the importance of demonstration quality for prompted translation [6], while Moslem et al. investigate adaptation

using in-domain examples and terminology [7]. These studies motivate assessing translation capability under explicitly defined conditioning rather than assuming uniform performance across languages and domains.

Translation benchmarks operationalize these comparisons. FLORES-101 provides professionally translated, fully aligned multilingual references for many-to-many evaluation [8]. WMT25 compares submitted systems, LLMs, and online providers using challenging test material, multiple domains, and document context [9]. Its professional evaluation covers 15 of the 30 offered language pairs using Error Span Annotation or Multidimensional Quality Metrics (MQM). This work underscores the importance of test difficulty, language coverage, and evaluation context; our segment-level benchmark does not replicate its document-level protocol.

Model comparison requires complementary quality measures. BLEU measures reference n-gram overlap [1], SacreBLEU standardizes its reporting [2], and chrF++ combines character and word overlap [10]. COMET learns quality prediction from human judgments [3], whereas BERTScore compares contextual token representations [4]. Expert MQM evaluation demonstrates the importance of annotator expertise and document context [11]. Terminology accuracy supplies an additional dimension, as reflected in dedicated WMT evaluation [12]; variation-aware analysis further distinguishes legitimate alternatives from undesirable inconsistency [13]. Reference-term retention, however, is narrower than bilingual terminology accuracy: span presence does not establish contextual correctness, and reference mismatch need not indicate mistranslation.

Terminology extraction supports the annotations needed for these diagnostics. Chun et al. use syntactically retrieved demonstrations to improve LLM term-boundary prediction on specialized benchmarks [14]. Such work motivates evaluating extraction quality independently: verifying occurrence in a reference does not establish annotation completeness or expert validity.

In chemistry, related tasks must be distinguished from prose translation. STOUT studies molecular representation-to-name conversion with structure-oriented checks [15], while CHEMDNER evaluates chemical mention recognition against expert annotations [16]; neither benchmarks multilingual chemical prose translation. For legal comparison, JRC-Acquis [17], distributed through OPUS [18], provides parallel text. LegISTyr distinguishes terminology insertion from contextual choice and fluency [19], and an EEA terminology suite reports performance that need not follow general MT rankings [20]. Our study centers on multilingual chemical-patent translation capability, using legal text as a comparison domain and terminology metrics as complementary diagnostics. Reference-derived terminology hints constitute a separate oracle-assisted condition, not evidence of source-only capability.

3 Data and Methodology

3.1 Evaluation Data and Preparation

Google Patents Chemical. The source is the BASF-AI within-document abstract-pairs dataset, documented locally as within-publication abstract pairs from the CLIR Google Patents Chemical corpus [21]. The exporter normalizes whitespace, rejects empty text, retains accepted upstream judge verdicts labelled *exact* or *high*, and selects in deterministic dataset order. These are alignment-screening labels, not human chemical validation. The snapshot contains 2,750 pairs in 11 stored directions, capped at 250 per direction. Both-side approximate lengths of 128–384 tokens are preferred, followed by source-only, target-only, and out-of-range backfill; the respective counts are 1,521, 626, 366, and 237. Length is a preference, not a hard criterion.

The active chemistry build retains German, English, Spanish, French, Russian, and Chinese, omitting Japanese. Nine stored directions plus their source/target swaps yield 18 evaluation directions and 4,500 planned segments. Reversal does not create independent publications. The upstream split name `train` denotes the input partition, not training of our translation systems.

The export has no pinned upstream revision, so the local snapshot is needed for exact reuse.

JRC-Acquis articles. The article snapshot uses OPUS JRC-Acquis v3.0 aligned Moses files [17, 18]. It selects 250 document anchors shared across English, Spanish, German, French, and Portuguese, generating all 20 ordered directions: 5,000 pairs, 250 per direction. Reverse rows are exact swaps; shared anchors do not imply identical passage boundaries across every unordered pair. The separate definition snapshot is not included here.

After legacy markup cleanup and whitespace normalization, usable aligned units contain words, 3–180 approximate tokens on each side, and a maximum length ratio of 3. Units are concatenated within document boundaries, targeting 450 source tokens with nominal bounds of 250–700 and at most one chunk per document per pair. Strict filtering rejects residual markup, controls, all-capital noise, and incomplete boundaries; it does not identify languages automatically. Recorded article source lengths are 251–679, mean 418.6 approximate tokens, not 256. Counts use the repository whitespace counter, not a provider tokenizer. This approximation is especially limited for Chinese; the task remains segment-level despite this larger snapshot.

Partitioning and overlap. Both resources are evaluation inputs; no fine-tuning or finalized train/development/test partition is specified. Small anchor pilots are pipeline checks, not independent held-out evidence. Before final scoring, prompt-development and evaluation examples must be separated by JRC anchor and patent publication, preferably family. Identifier and normalized-text duplicate audits, including reverse directions and pilot overlap, remain pending. Public-corpus inclusion in proprietary pretraining cannot be excluded; preserved identifiers facilitate audits but do not establish contamination-free data.

3.2 Reference Terminology Annotation

Four automatic approaches operate on target references: domain-prompted LLM span proposals, Stanza/Universal Dependencies syntactic spans, XLM-R/NOBI token-classified terms, and spaCy entities, noun chunks, and filtered n-grams. Normalized duplicate keys merge provenance while retaining surface forms; LLM proposals absent from the reference are rejected. The article configuration also imports candidates from an earlier GLM-labelled manifest. This is reused automatic evidence, not human annotation, and must be controlled in extractor ablations.

Chemistry attaches evidence from local IATE, Wikidata, and AGROVOC; articles use local IATE, Wikidata, and UNTERM. These respectively supply EU terminology, multilingual labels/aliases, FAO controlled vocabulary, and official UN terminology. [1] Scope and coverage differ: UNTERM's official languages exclude German and Portuguese. Local IATE uses a CSV-derived SQLite index. Matches, variants, identifiers, and categories are stored as evidence. Lookup misses or timeouts do not establish invalidity; chemistry network lookups have a configured one-second timeout.

Domain-specific LLM curation selects numbered, enriched candidate IDs given the reference; accepted terms remain exact reference spans. Both active configurations cap candidates at 80 and refined terms at 16 per segment. They record `gpt-6-luna`, Responses API, temperature 0, 1,024 output tokens, disabled thinking, and reasoning effort `none`; this endpoint identifier is not an independently verified immutable release. Chemistry uses four workers and a 180-second LLM timeout; articles use two workers and 120 seconds. Manifests retain extractor provenance, external support (`verified_by`), and curation separately. Externally supported, LLM-selected, and human-validated are distinct statuses; no expert gold terminology set or chemical-structure validation is implemented. Chemical notation is preserved as text, not validated as molecular identity.

[1] https://iate.europa.eu/; https://www.wikidata.org/wiki/Wikidata:Introduction; https://www.fao.org/agrovoc/about; https://unterm.un.org/unterm2/.

3.3 Translation Generation

The one-shot translator sends one segment, language labels, and a chemistry or legal prompt to the selected provider, requesting only a translation and preservation of terms, formulas, identifiers, quantities, units, and legal effect as appropriate. There is no fine-tuning, retrieval augmentation, or constrained decoding. The planned paired comparison holds model and decoding settings fixed while toggling manifest-term injection. A source-only baseline sees no reference information. The other condition adds up to eight refined target terms with instructions to use them only when the concept is present: soft hints, not enforced lexical constraints.

Since these hints derive from the evaluation reference, this is explicitly an *oracle terminology* condition, not deployable source-only glossary induction. Pilot configurations name `gpt-4.1-nano` and `gpt-4.1-mini`; a legal run also names `gpt-6-luna`. The final cross-domain roster and provider versions remain unspecified. Temperature 0 does not guarantee deterministic hosted inference. Predictions, supplied terms, model metadata, and failures are saved as JSONL. Dry-run source copying is a pipeline check, not an MT baseline.

3.4 Quality and Terminology Evaluation

The primary reporting set will combine corpus SacreBLEU and chrF++, segment COMET and BERTScore, and strict/variant-aware target coverage. BLEU uses four word n-gram orders, case-sensitive `13a` tokenization, and corpus scoring [1, 2]; chrF++ uses character order 6, word order 2, and $\beta = 2$ [10]. They require hypotheses and references, reward surface agreement, and can miss consequential chemical substitutions. Effective-order sentence BLEU is diagnostic, not a substitute for corpus BLEU. This is not an official WMT25 scoring-protocol reproduction.

COMET uses source/hypothesis/reference with `Unbabel/wmt22-comet-da` [3]; BERTScore uses hypothesis/reference contextual token alignment with `xlm-roberta-large`, recording precision, recall, and F1 [4]. These are computed per segment and averaged. BERTScore precision/F1 concern embedding alignment, not chemical-term precision/F1; neither metric certifies chemical identity, formula correctness, or legal effect.

For segment i, let T_i contain applicable annotated target terms and $c_z(t)$ count occurrences of t in text z. Implemented strict coverage is

$$C_i = \frac{100}{|T_i|} \sum_{t \in T_i} \min\left(\frac{c_{h_i}(t)}{c_{r_i}(t)}, 1\right), \quad c_{r_i}(t) > 0, \tag{1}$$

with hypothesis h_i and reference r_i. Variant-aware coverage accepts explicit stored variants in the numerator. Counting uses Unicode NFKC, case folding, whitespace normalization, and substring matching, not word-boundary or chemical-entity matching. Case folding can erase meaningful chemical case distinctions; extra hallucinated terms are not penalized. Report refined-term coverage and its externally supported subset separately. Segments with no applicable terms are excluded, with counts reported. Source-conditioned terminology success requires source-term alignment; target-only annotations do not establish bilingual term accuracy.

Term-only BERTScore recall aligns the hypothesis against concatenated reference terms as a soft diagnostic. The registry additionally supports reference-free COMETKiwi, XCOMET error spans, and a domain-specific LLM MQM-style judge. The judge returns a 0–100 score and minor/major/critical counts with weights 1/2/5; it is automatic, not expert annotation or an exact WMT FSP reproduction. These remain supplementary pending access and calibration. BLEURT is optional and checkpoint-dependent; NIST and terminology precision/F1 are not selected measures.

3.5 Experimental Protocol and Reproducibility

Report domains and directions separately, macro-averaging direction means and recomputing corpus overlap rather than averaging sentence BLEU. Paired comparisons use the same segments

under baseline and oracle prompts. Proposed ablations isolate extractor families, external evidence, and curation, retaining exact-span validation. Annotation yield is not extraction accuracy without human labels; cross-segment consistency and expert error analysis are not yet measured. Proposed paired resampling groups by document/anchor and retains reversals together; statistical testing remains unimplemented.

Final scoring requires freezing the partition, model roster, metric subset, aggregation, and uncertainty procedure. Reproduction artifacts include source checksums, Git revision, locked software, extractor checkpoints/language models, exact prompts, provider identifiers and decoding settings, dated IATE exports, and lookup caches. Selection, extraction, enrichment, curation, and manifest checkpoints support stage-scoped hash reuse. Authentication/network failures are not completed examples; successful-row denominators and failure counts must remain visible.

4 Results and Analysis

Comparative translation results are not yet available. Planned analysis contrasts baseline and oracle conditions by domain and direction, general quality with term coverage, and chemical-name, formula, quantity, omission, and legal-effect errors. No empirical findings are claimed.

5 Conclusion

The methodology links bounded parallel segments to auditable reference terminology and complementary evaluation measures. Its empirical value, oracle-term effects, and cross-domain generality remain to be established by the planned comparisons.

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